**SE375 SYSTEM PROGRAMMING SPRING 2025-2026**
**Laboratory Assignment**

**May 15, 2026**

# 1. Introduction

Assume that each district (Konak, Gaziemir etc) of İzmir reports their data on public transport usage and the overall data set is collected for analysis. The data set from each district contains the daily number of bus passengers and number of bus trips. An example view of a data set from one district (say Konak) is as following;

| Date | Day of the Week | Number of Trips | Number of Passengers |
|---|---|---|---|
| 01.02.2023 | Wednesday | 34 | 1242 |
| 02.02.2023 | Thursday | 51 | 4589 |
| 03.02.2023 | Friday | 30 | 2568 |
| 04.02.2023 | Saturday | 48 | 3874 |
| 05.02.2023 | Sunday | 36 | 2514 |
| 06.02.2023 | Monday | 44 | 6257 |
| 07.02.2023 | Tuesday | 27 | 2674 |

# 2. Task

In this lab assignment, you are going to develop a Java Application that will calculate statistics on given data. Your application will read each data file, compute and store the values and report to the user.

## a. Specifications

- Each data file contains the daily number of bus trips and number of bus passengers in a CSV (comma-separated values) format for each district. The first line of the file has the headers in the order of:

  *Date*, *Day of the Week*, *Number of Trips* and *Number of Passengers*

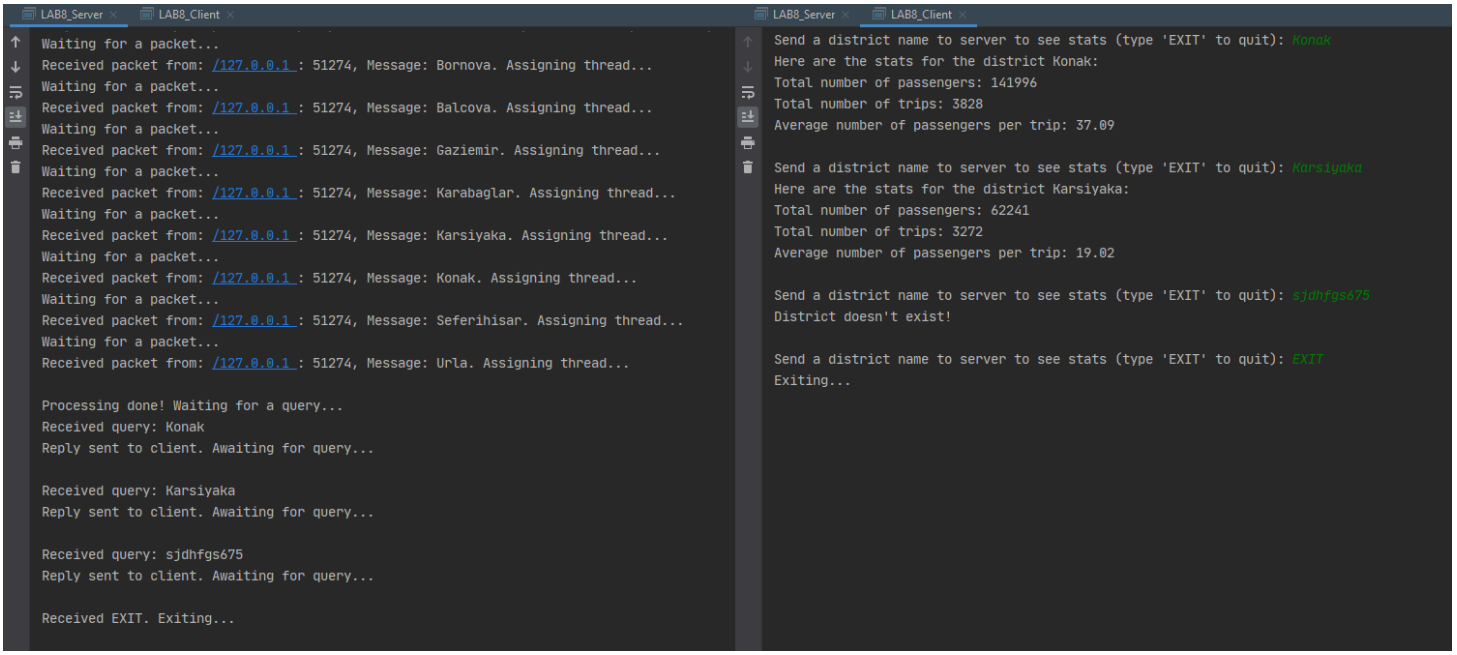
  and the remaining lines include data for each date of the month.
- `DD.MM.YYYY` is the date format.
- Day of the week is one of the below;

*Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday*

- The data type of *Number of Trips* and *Number of Passengers* is integer. It is guaranteed that the data files do not have missing values.
- Your application should consist of two parties: multiple **clients**, and a **server**.
    - You should have at least two .java files, one for each party, each with its own application entry point (a main method).
- All data communication should proceed using a TCP socket.
- The **client** should **send** the **server** the names of all data files (e.g. “Balcova.csv”, “Karsiyaka.csv”).
    - We can assume that both the **client** and the **server** are aware that there are 8 files to send in total.
- The **server** should **display** the address and port of the sender and the file names.
- The server should **process** the files, and **create** a `HashMap` of the result, where:
    - The keys are district names,
    - The values are total number of passengers, total number of trips and average number of passenger per trip.
    - The server should work in a multi-threaded fashion, where each thread should process a single csv file.
        - For synchronization, the server should make use of the Lock functionality.
- The **client** should then **query** the server by **sending** it the name of a district (e.g. “Balcova”).
- The **server** should, after **receiving** the query, **display** it, and **send back** the **client** the corresponding statistics. The **client** should **display** the results on its end.
    - If the query is incorrect, it should **send** an error status to the **client**. This error could be a `String` message, or an invalid numeric value (e.g. negative value).
    - You will need to implement conversions between numeric data types and byte arrays.
- This procedure should be repeated until the **client sends** the query message “EXIT”, in which case both the **client** and the **server** should be **terminated**.

## b. Example Runs

Here are example console output screenshots, both from the server side and the client side:




# 3. Regulations

- You should use Java.
- Late submission is **not** allowed.
- This is an **individual** assignment. Plagiarism, copying, cheating, outsourcing the exam to another person or organization for with or without pay are considered as actions of academic dishonesty. Failure to maintain academic honesty may result in disciplinary action according to the Izmir University of Economics' disciplinary bylaw for students of institutions of higher education ( https://www.ieu.edu.tr/en/bylaws/type/read/id/13 ).
- Submissions should be done to Blackboard. Submissions via e-mail will not be accepted.
- You should make sure to obey the specifications.

# 4. Submission

1. You should submit 2 .java files to Blackboard containing your Java files.
2. By submitting you pledge the following; *"On my honor, as an Izmir University of Economics student, I affirm that I have not given or received any unauthorized help on this exam, and that this work is my own."*